Models with Multiple Continuous Predictors.

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Multi-variable Regression

- Usually, conditioning our predictions on a single variable won't be enough.
  - This is because most variables we might measure (height, academic achievement, probability of winning a match) are caused by multiple factors or at least associated with multiple factors.

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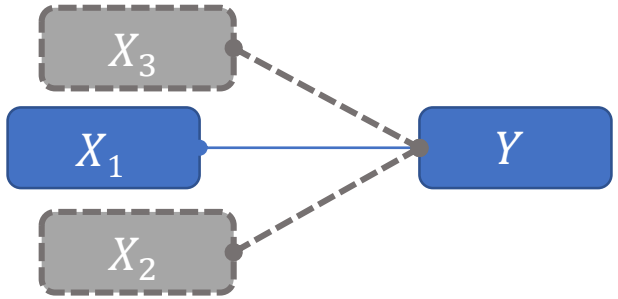

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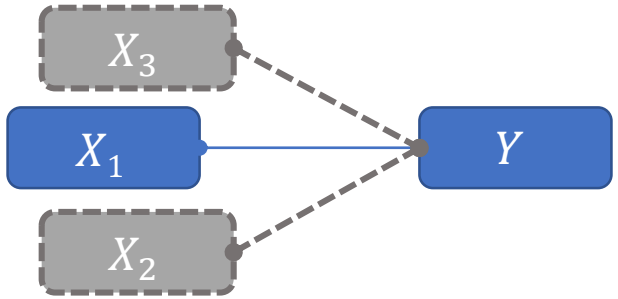
If X_1 is the only thing that causes Y, any e_i in our data would be due to measurement error.



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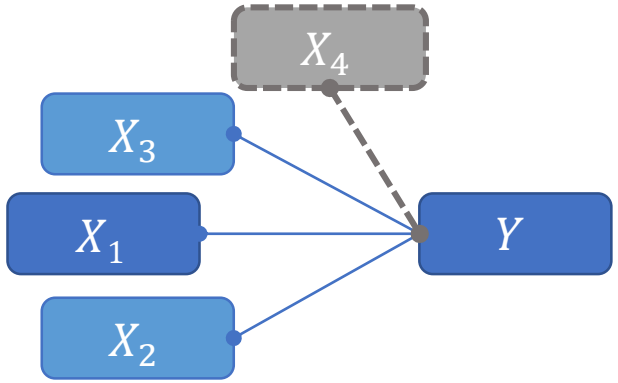
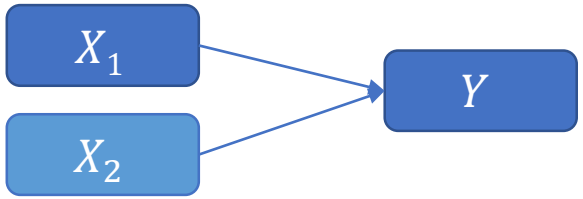


Adding these variables can reduce e_i, but there are likely still variables we haven't accounted for.

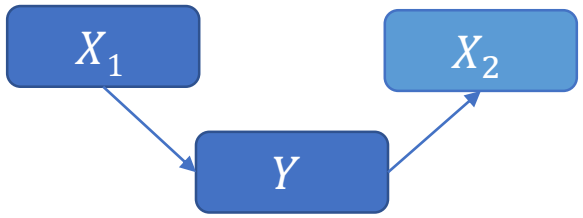
Multi-variable Regression

- Further, ***causation*** is not established by the statistical relationship but by logical analysis/experimental design.

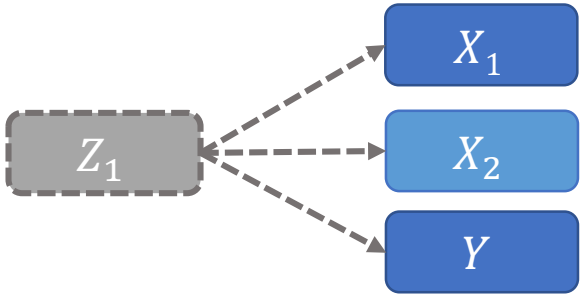
Some example associations:



X_1 and X_2 both cause Y.



X_1 causes Y, which in turn causes X_2.

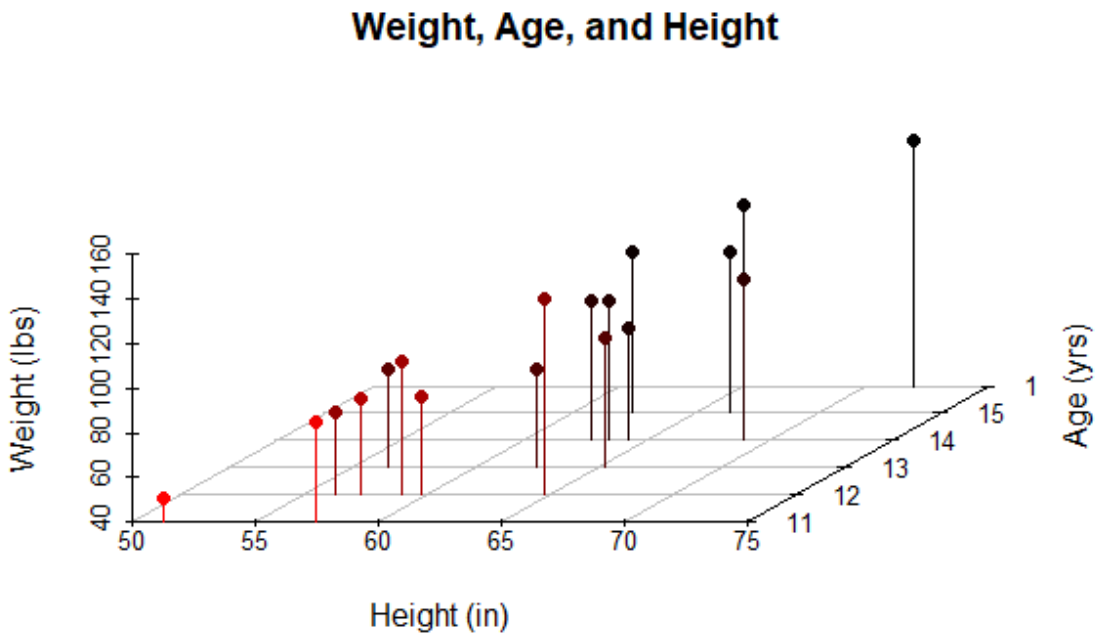


X_1, X_2, and Y are all caused by an unmeasured third variable Z_1.

Unpacking a multi-variable relationship

- For this example, we will look at the relationship between students' heights, weights, and ages presented in the chapter.

```
> head(dat)
     Name Sex Age Height Weight
1  Alfred   M  14   69.0  112.5
2   Alice   F  13   56.5   84.0
3 Barbara   F  13   65.3   98.0
4   Carol   F  14   62.8  102.5
5   Henry   M  14   63.5  102.5
6   James   M  12   57.3   83.0
```



Going from univariable to multivariable regression:

- Multivariable model:

$$\hat{y}_i = b_0 + b_1(x_{1i}) + b_2(x_{2i})$$

Intercept

The effect of X_1 on Y_P

The effect of X_2 on Y_P

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Going from univariable to multivariable regression:

- How do we interpret a single variable in the multivariable model?

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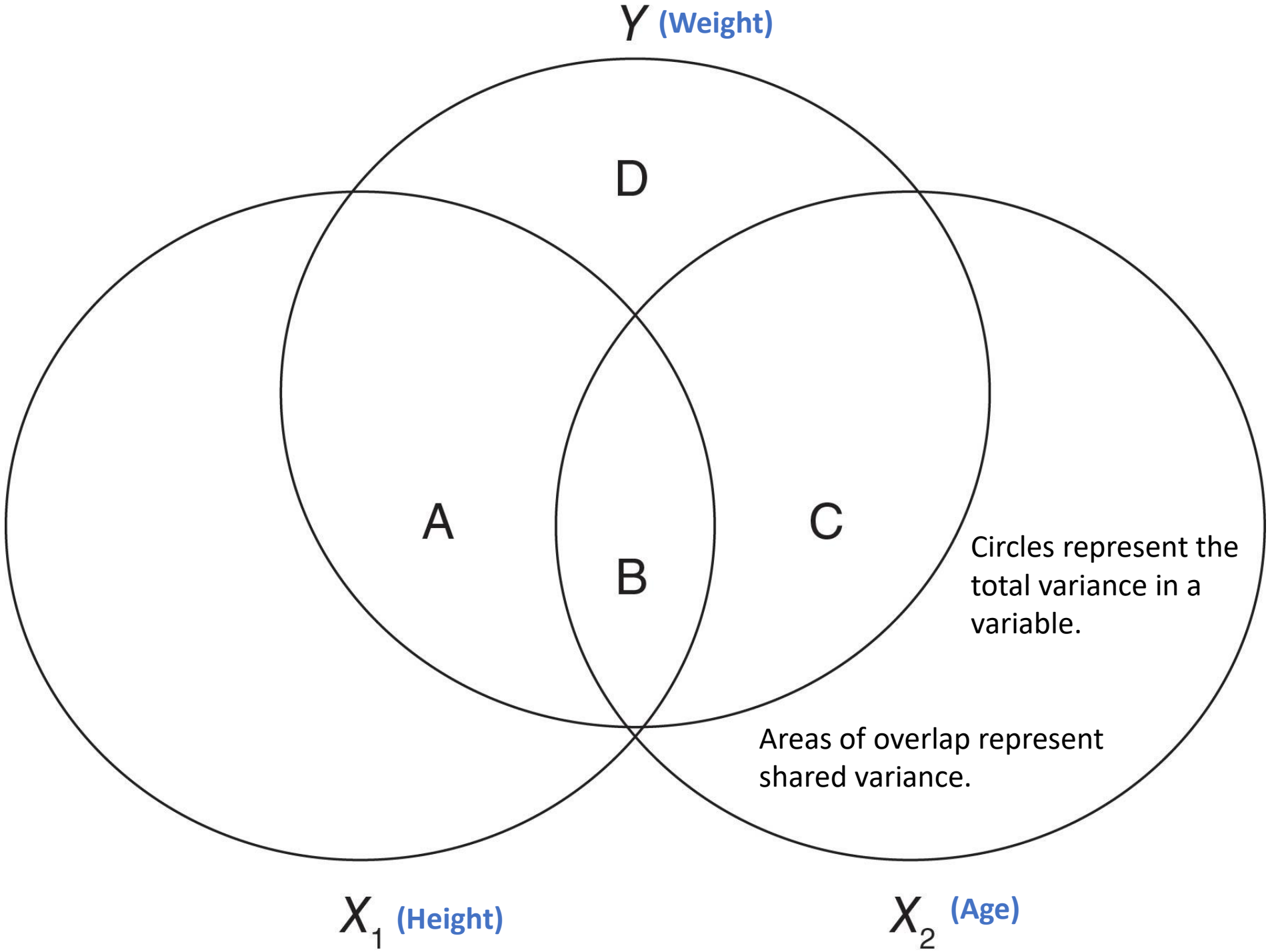
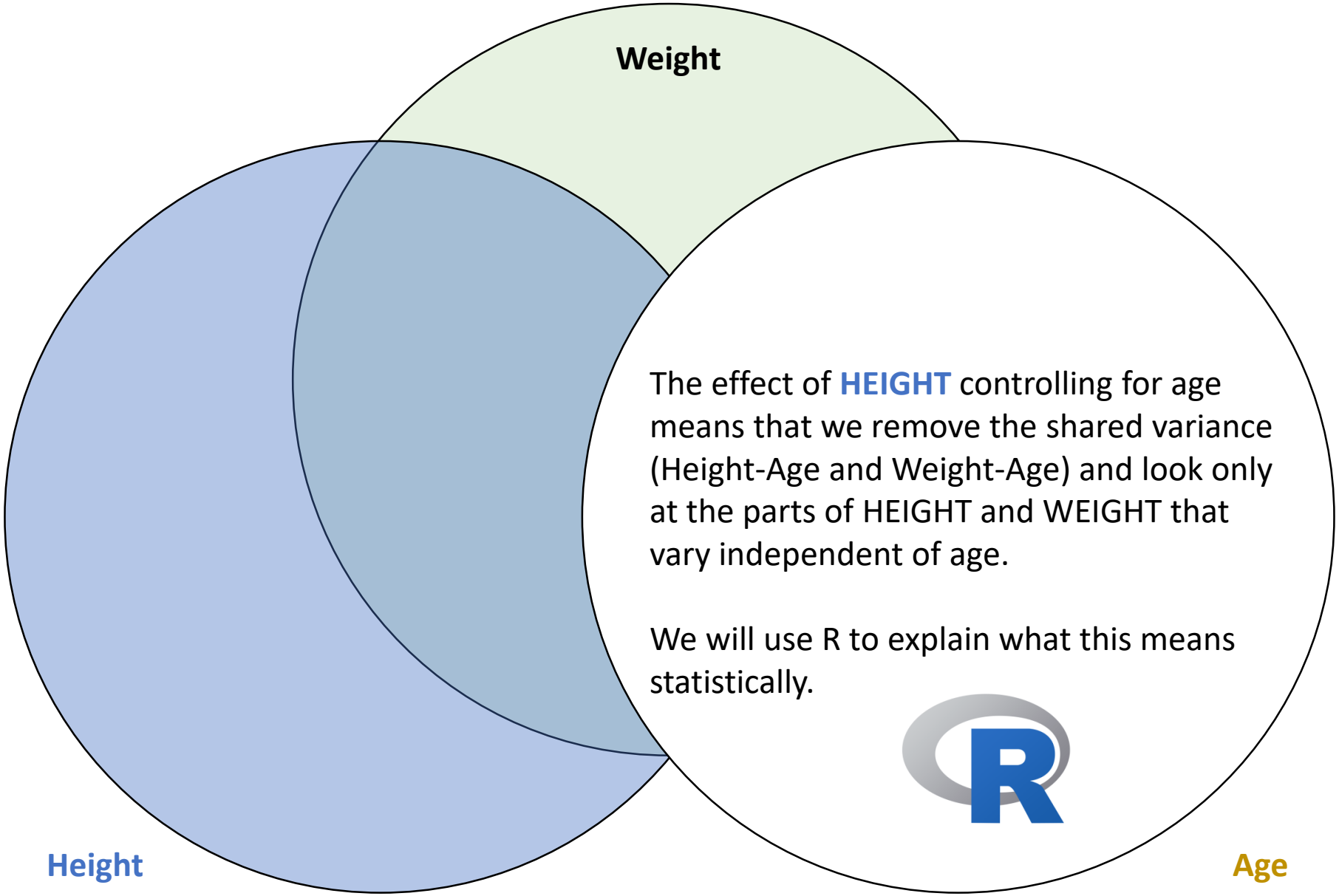
Similarly, this is the effect of Age controlling for Height, but what does it mean to control for height?

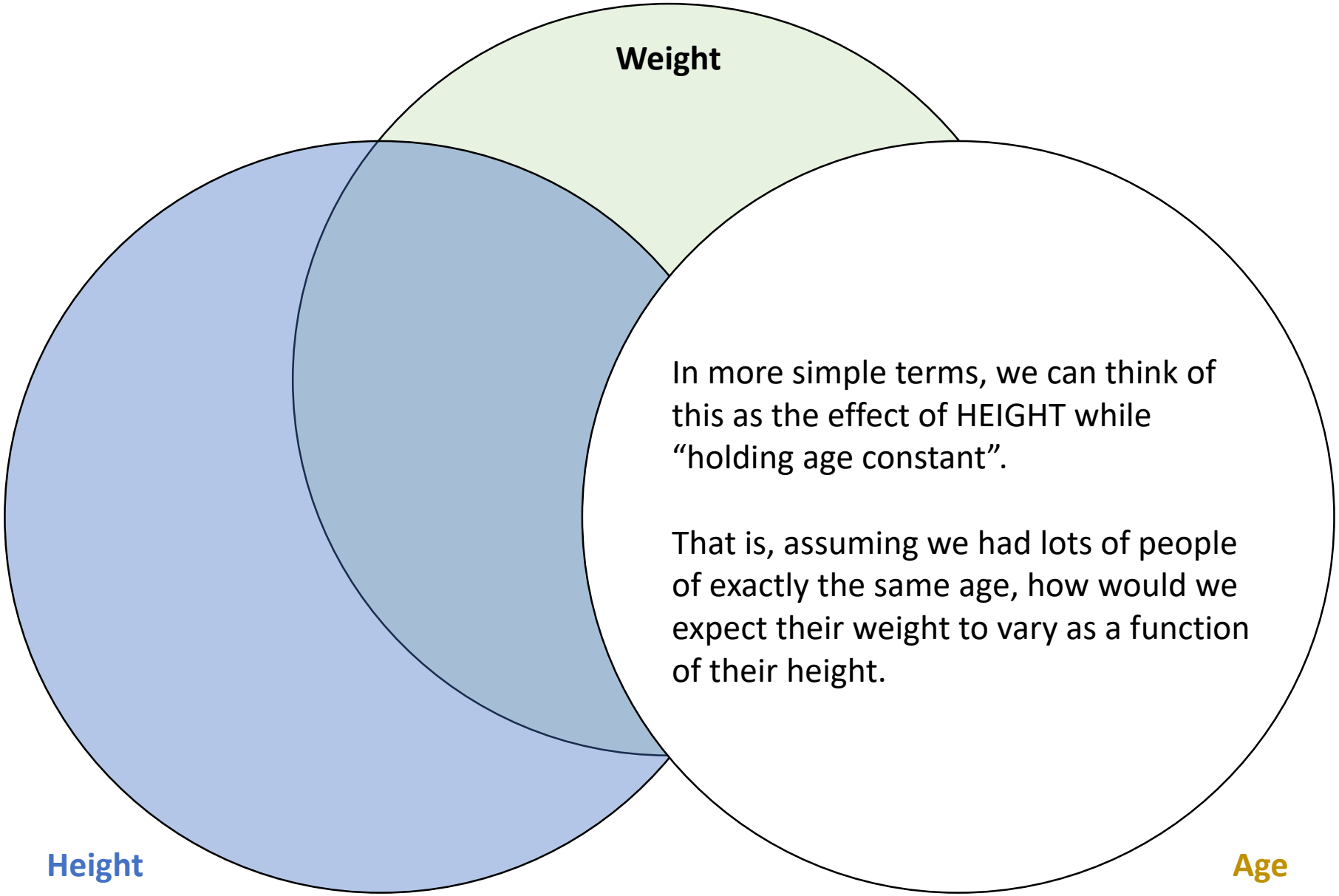
Controlling for Shared Variance


Y (Weight)
D
A
B
C
Circles represent the total variance in a variable.
Areas of overlap represent shared variance.
X1 (Height)
X2 (Age)



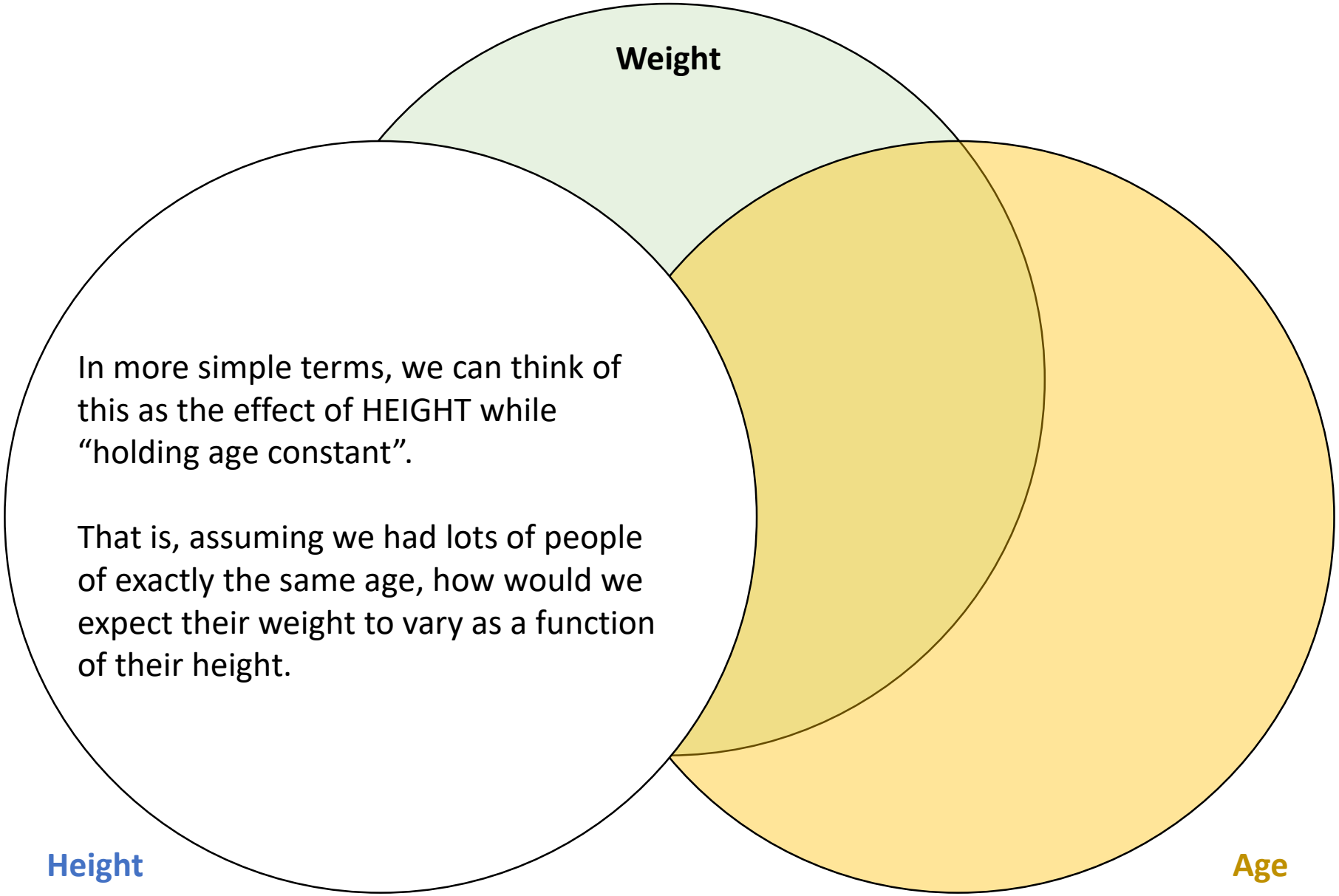
The effect of **HEIGHT** controlling for age means that we remove the shared variance (Height-Age and Weight-Age) and look only at the parts of HEIGHT and WEIGHT that vary independent of age.

We will use R to explain what this means statistically.



In more simple terms, we can think of this as the effect of HEIGHT while "holding age constant".

That is, assuming we had lots of people of exactly the same age, how would we expect their weight to vary as a function of their height.



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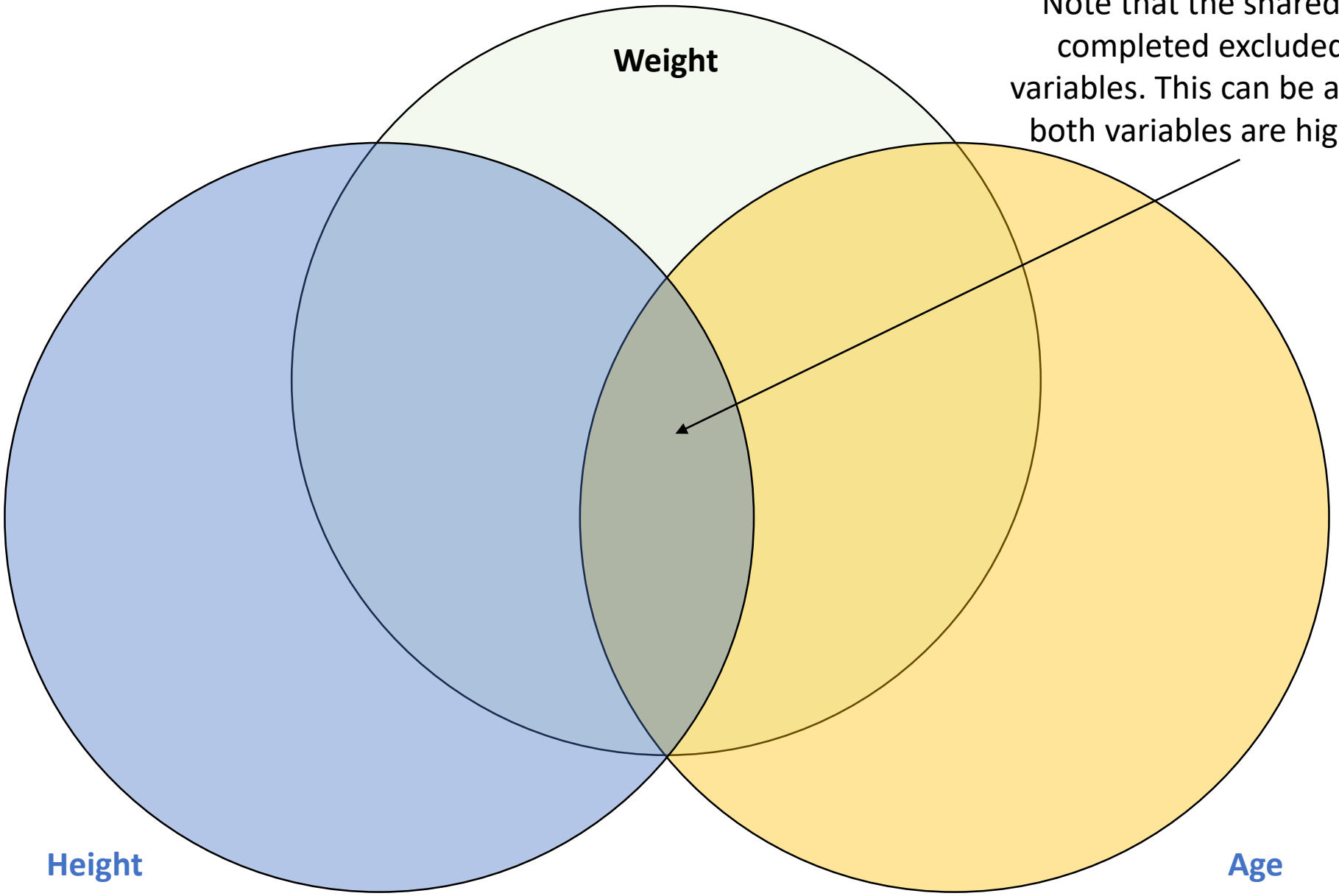
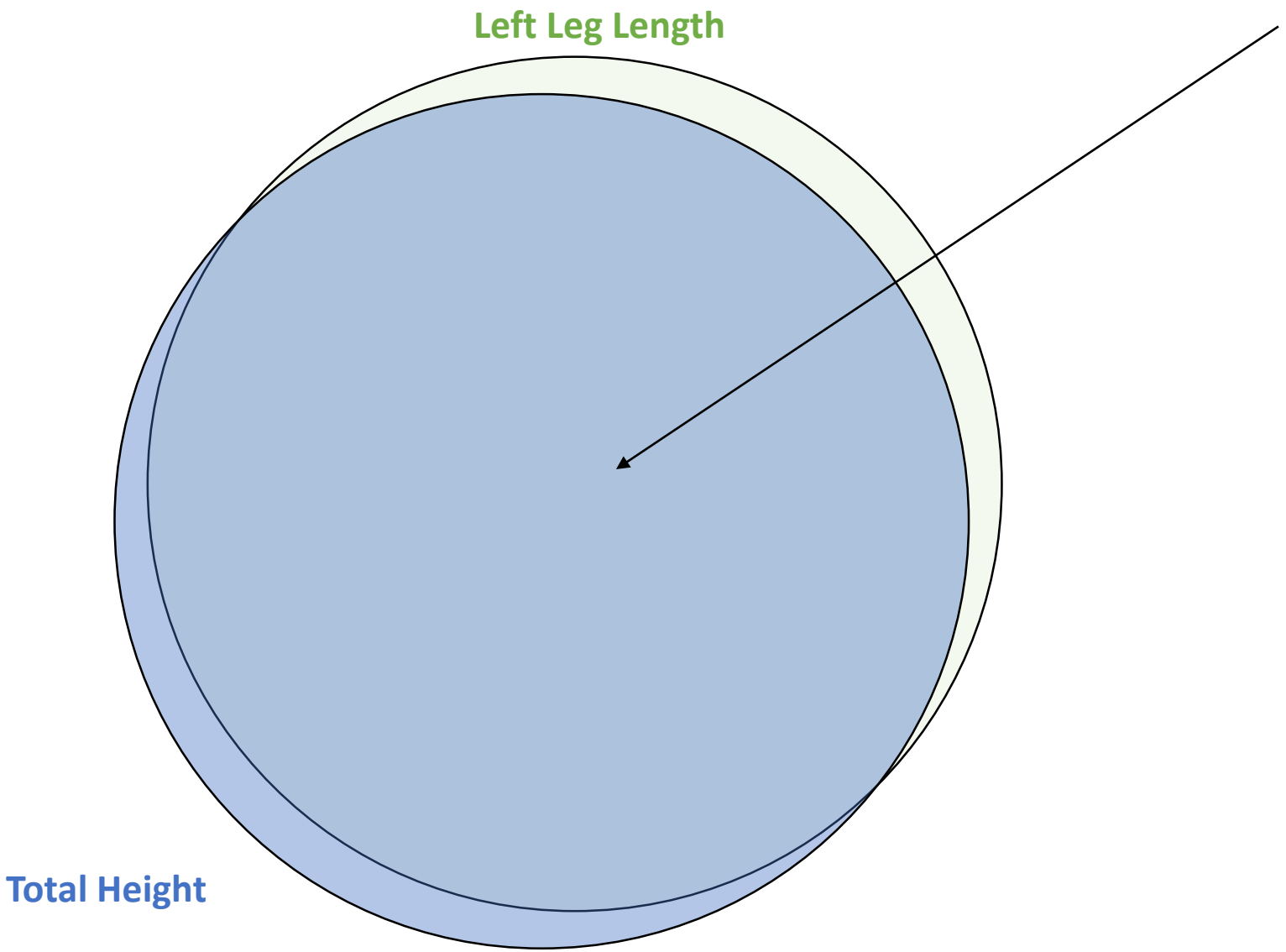


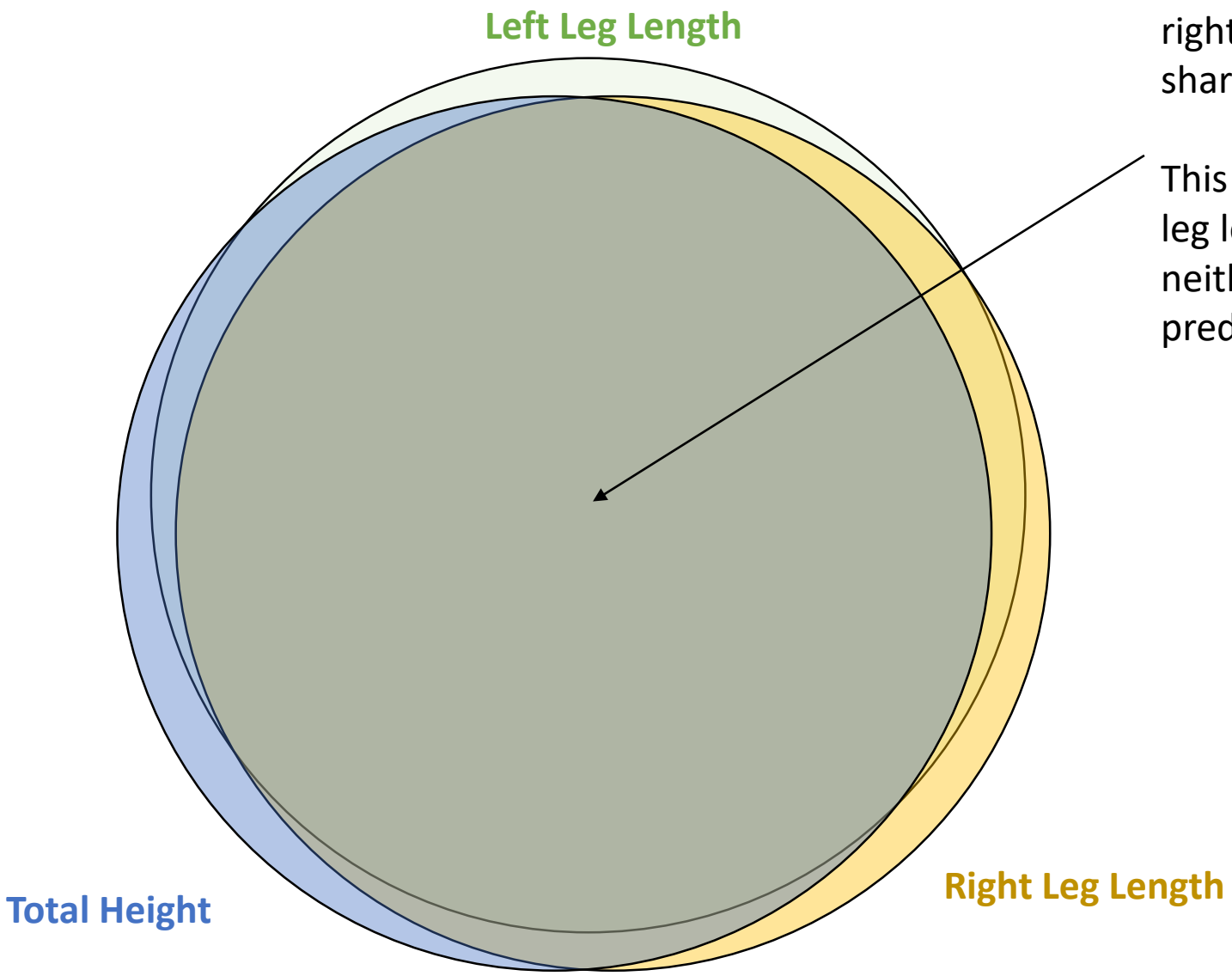
Note that the shared variance is completed excluded from both variables. This can be a problem is both variables are highly related.

Collinearity



On it's own, the length of the left of a person's left leg is a very good predictor of their total height.

Collinearity



However, if we put both left leg length and right leg length in the same model, their shared variance will be excluded.

This collinearity (i.e., correlation) between left leg length and right leg length means that neither of these variables is a very good predictor controlling for the other.